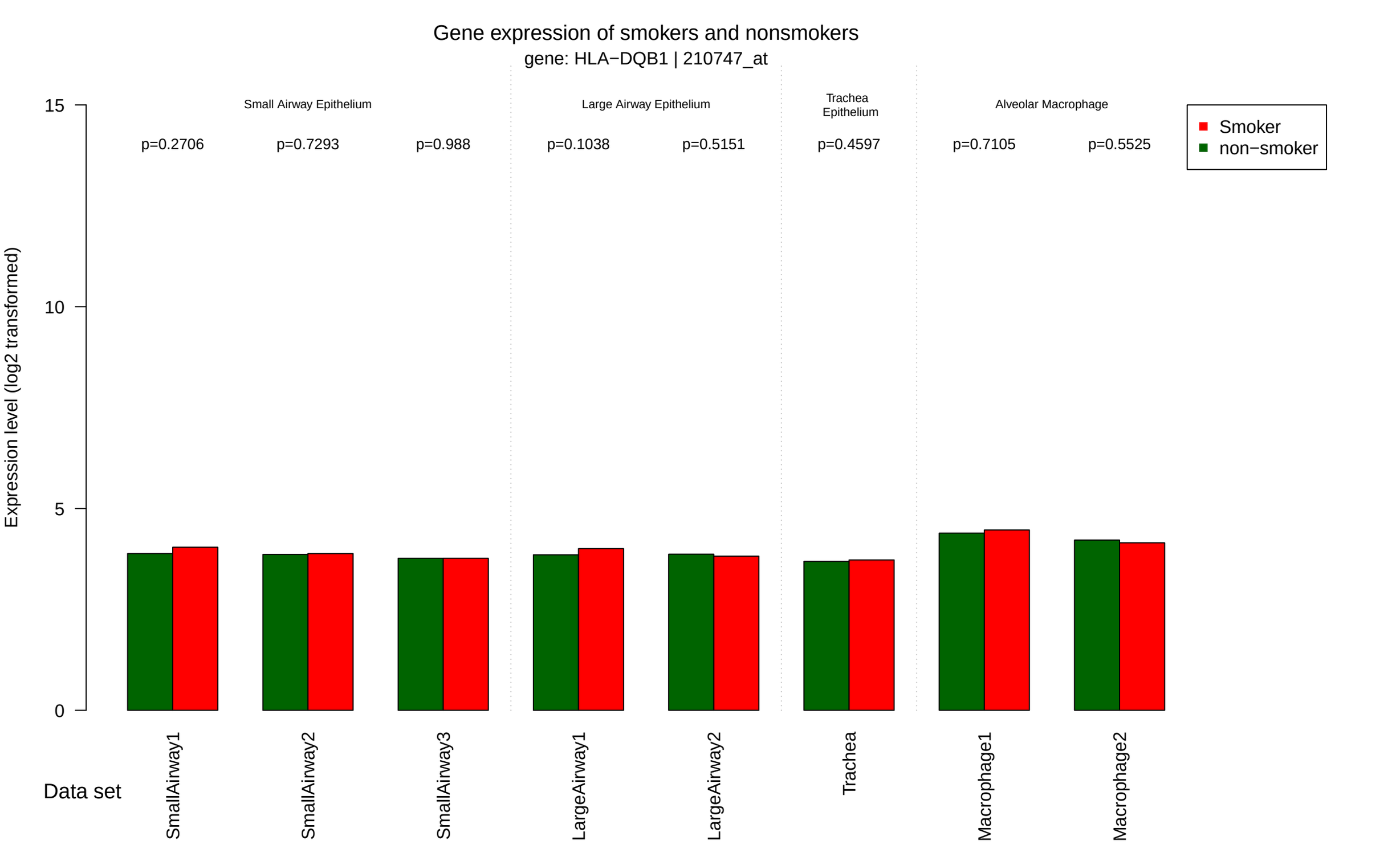
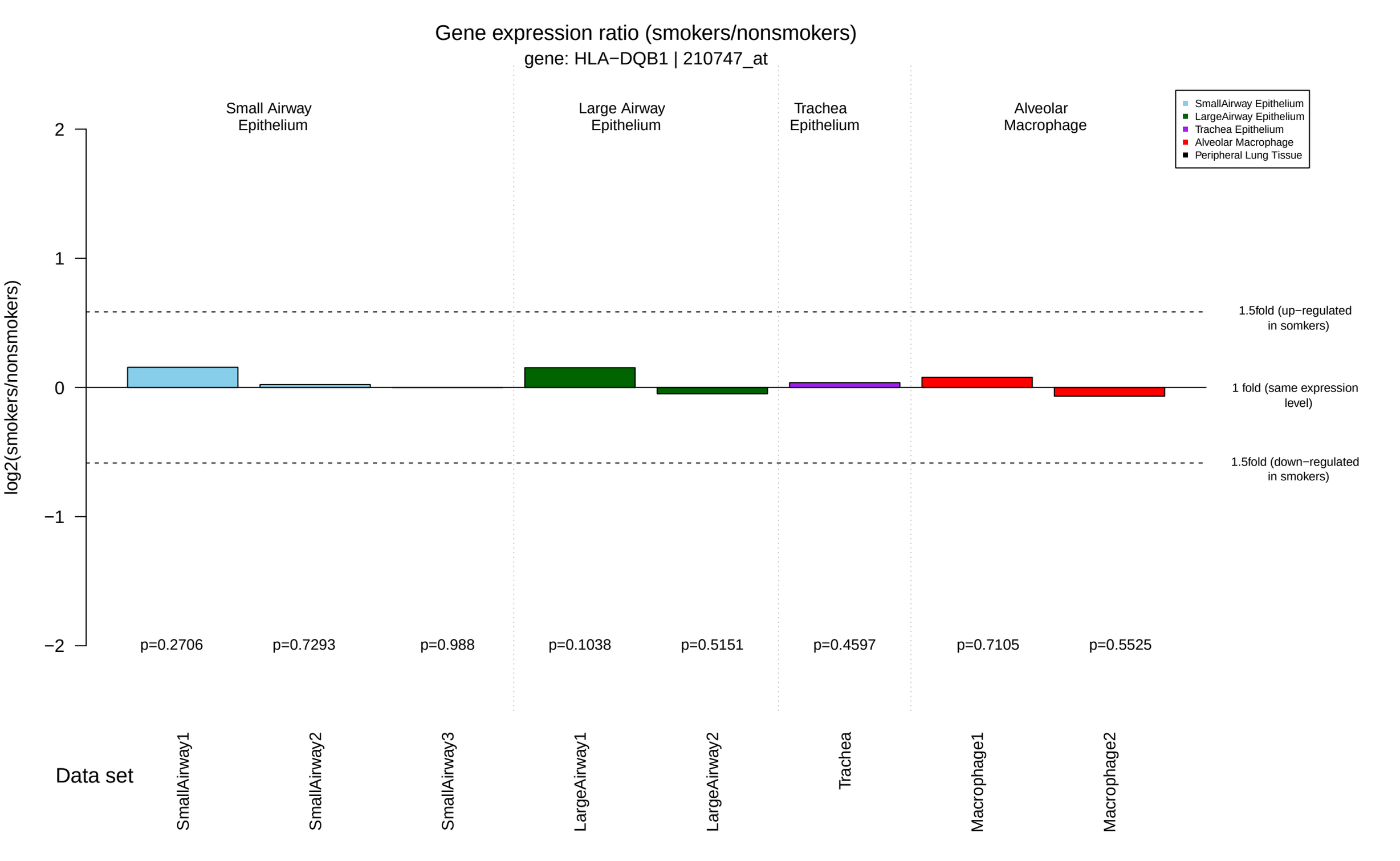
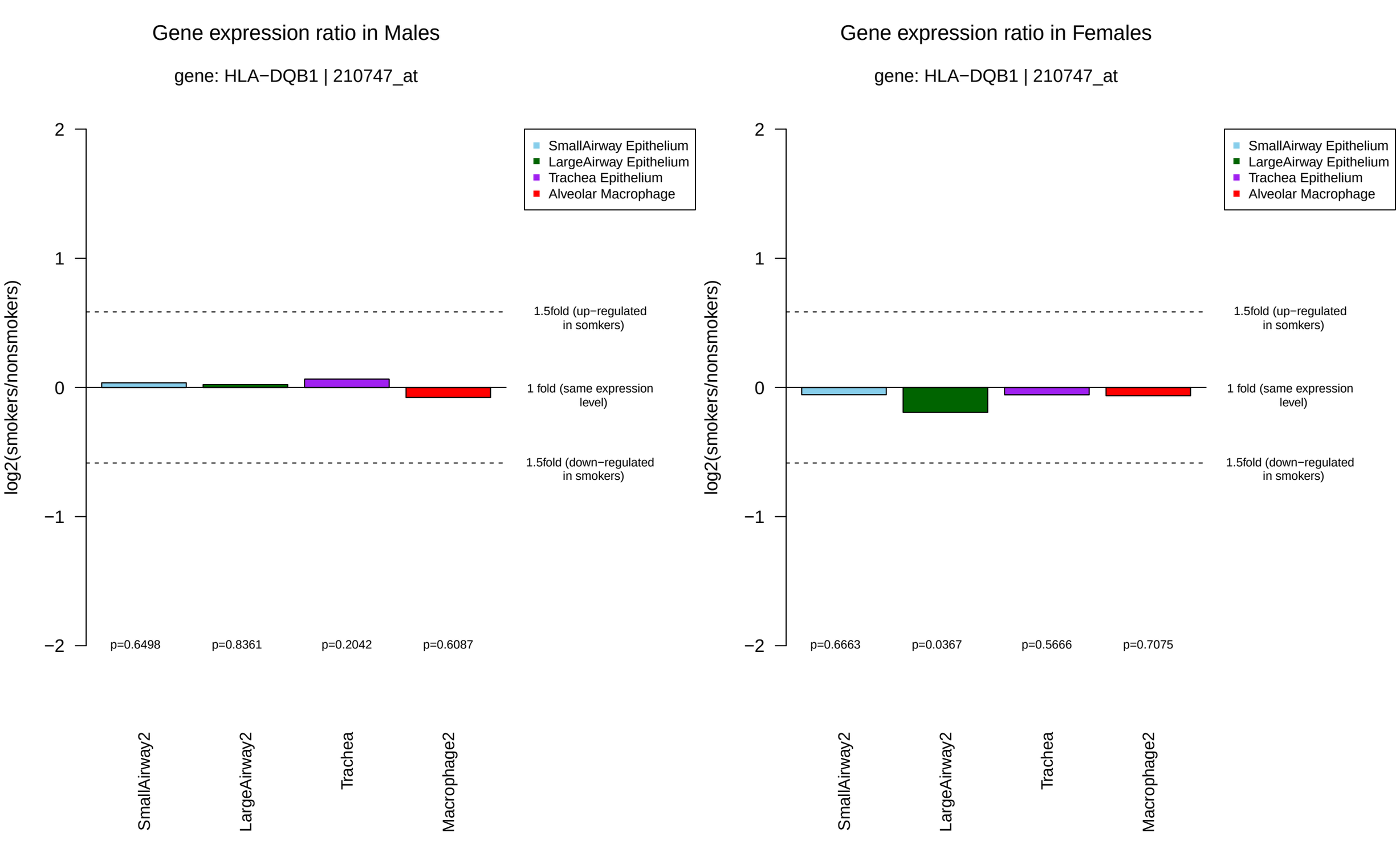
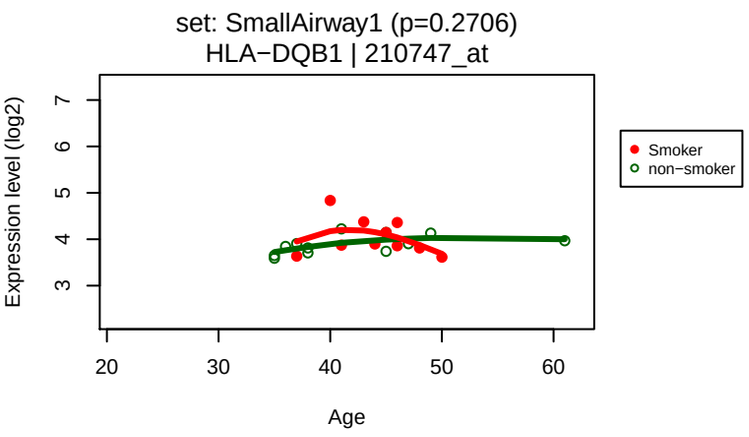
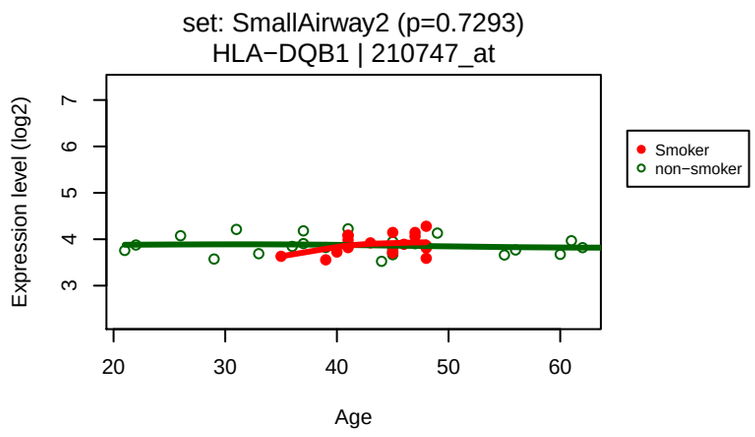
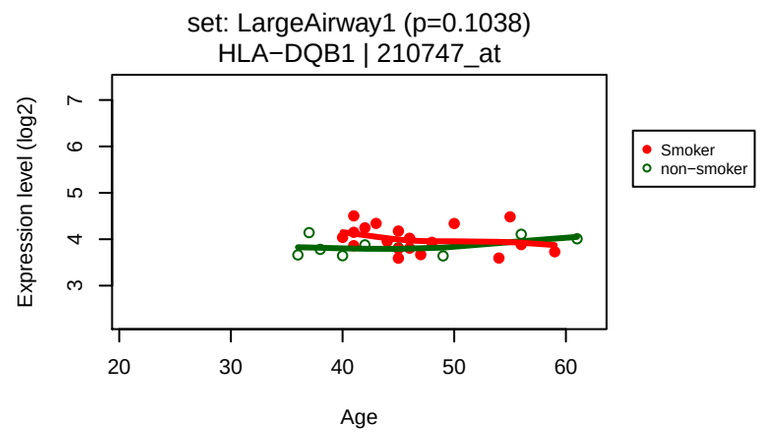

Gene expression of smokers and nonsmokers
gene: HLA−DQB1 | 210747_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2706
p=0.7293
p=0.988
p=0.1038
p=0.5151
p=0.4597
p=0.7105
p=0.5525
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

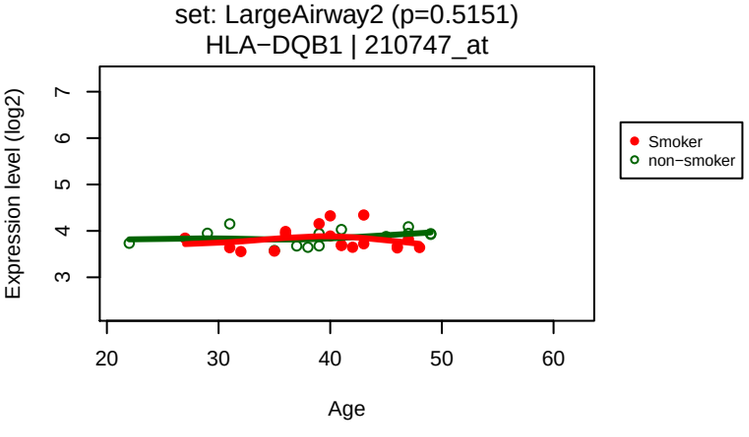

Gene expression ratio (smokers/nonsmokers)
gene: HLA−DQB1 | 210747_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2706
p=0.7293
p=0.988
p=0.1038
p=0.5151
p=0.4597
p=0.7105
p=0.5525
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

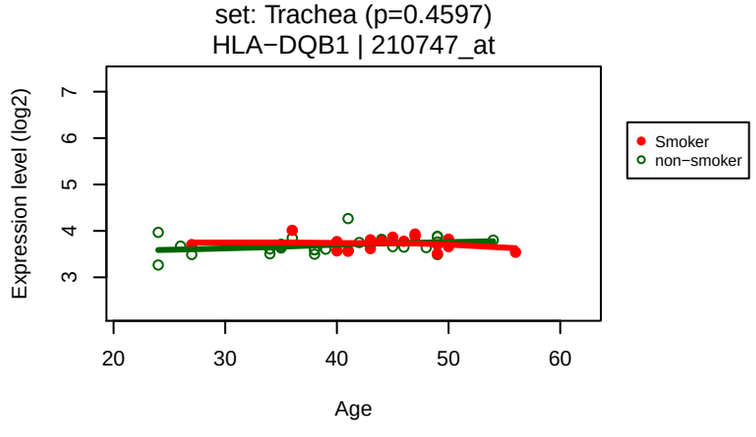

Gene expression ratio in Males
gene: HLA−DQB1 | 210747_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6498
p=0.8361
p=0.2042
p=0.6087
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HLA−DQB1 | 210747_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6663
p=0.0367
p=0.5666
p=0.7075
SmallAirway2
LargeAirway2
Trachea
Macrophage2

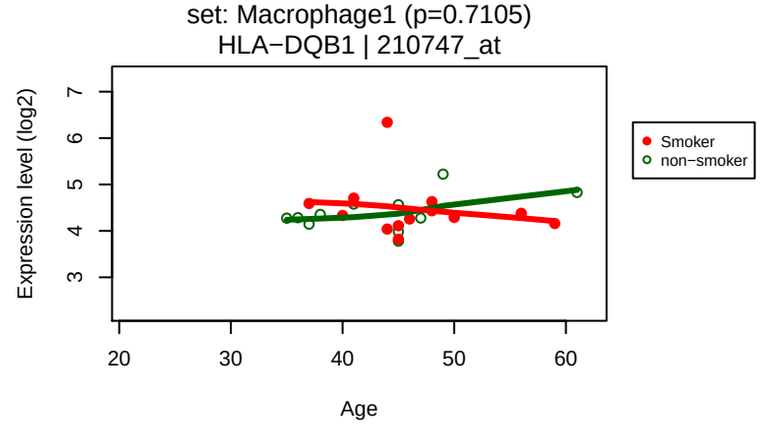

set: SmallAirway1 (p=0.2706)
HLA−DQB1 | 210747_at
Expression level (log2)
Age
Smoker
non−smoker

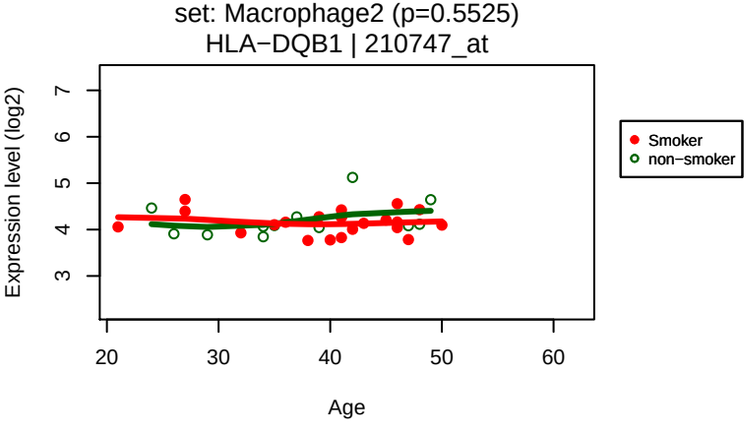

set: SmallAirway2 (p=0.7293)
HLA−DQB1 | 210747_at
Expression level (log2)
Age
Smoker
non−smoker

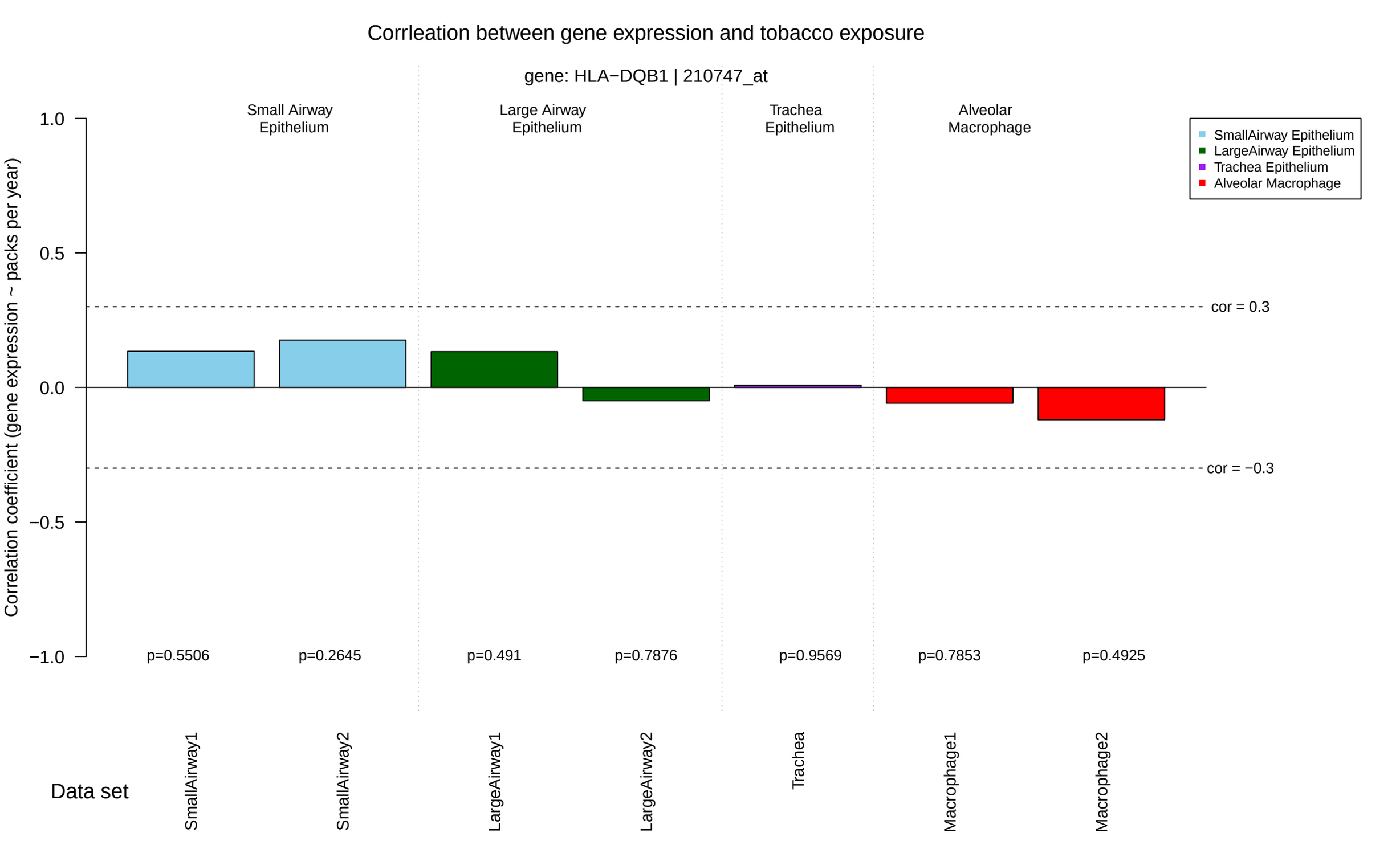

set: LargeAirway1 (p=0.1038)
HLA−DQB1 | 210747_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5151)
HLA−DQB1 | 210747_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4597)
HLA−DQB1 | 210747_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7105)
HLA−DQB1 | 210747_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5525)
HLA−DQB1 | 210747_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HLA−DQB1 | 210747_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5506
p=0.2645
p=0.491
p=0.7876
p=0.9569
p=0.7853
p=0.4925
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2